# 1.3 Solving One-Step Equations

Name _____ Period ____ Date _____

Algebra I 1.3 · Student sheet

**I can…**

- solve equations using addition or subtraction.
- solve equations using multiplication or division.
- write and solve equations for real-life problems.

**Vocabulary** · fill in the missing word

**Equation** — a statement that two _____ are equal

**Solution** — a value that makes the _____ true

**Inverse operations** — operations that _____ each other (+ and −, × and ÷)

**Properties of equality** — doing the same thing to both sides keeps an _____ true

## Part 1 · Undo it

| Equation | What was done to the variable? | Undo with | Solution | Check |
|---|---|---|---|---|
| $x - 7 = 12$ | | | | |
| $y + 5 = -2$ | | | | |
| $4n = -28$ | | | | |
| $\frac{m}{3} = 6$ | | | | |
| $k - 1.5 = -2.5$ | | | | |
| $(\frac{2}{3})p = 10$ | | | | |

## Part 2 · Notes: fill in the blanks

Inverse operations _____ each other: addition and _____; multiplication and _____.

Whatever you do to one side, do to the _____.

To undo multiplying by a fraction, multiply by its _____.

## Part 3 · Examples (work them with the class)

**Example 1**

Solve each equation.

a. $x - 6 = 11$ b. $w + 3.5 = 1$

### Example 2

Solve each equation.

a. $-7y = 42$ b. $\frac{t}{4} = -3$

### Example 3

Solve $(\frac{3}{4})x = 9$.

### Example 4

You buy 4 movie tickets for $46. Write and solve an equation to find the cost of one ticket.

## Part 4 · Partner practice

### Partner 1

Solve $n + 12 = 5$.

### Partner 2

Solve $-\frac{x}{5} = 2$.